Quantum Collapse as Petz Recovery Failure: An Information-Theoretic Resolution

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The measurement problem—why quantum superpositions give way to definite outcomes—persists as a central difficulty in the foundations of physics. We prove that wavefunction collapse is equivalent to the exponential failure of the Petz recovery map, the unique Bayesian retrodiction functor in quantum theory. Three results follow. (1) For an N-body composite, the total entropy production satisfies $\Sigma_{\text{total}} \geq N\,\sigma_{\min}$, so the recovery fidelity $F \leq \exp(-N\sigma_{\min}/2)$ decays exponentially in particle number; macroscopic objects classicalize without external observers (Theorem 1). (2) The Penrose collapse energy E_G equals $(c^4/32\pi G)\,\|\nabla(\Delta\Sigma)\|^2$ integrated over space, connecting objective reduction to a gradient instability of the entropy-production field Σ (Theorem 2). (3) The ghost-condensation kinetic function $K(Q) = \mu^2(Q-1)^2$ arises as the regularized quantum relative entropy of coherent-state excitations, identifying $\mu^2 = 2\pi$ with the de Sitter temperature and giving an information-geometric origin to ghost condensation (Theorem 3). The framework requires no modification of quantum mechanics and no new dynamical collapse law. We identify a category error in Penrose's argument for objective reduction—the conflation of pointwise metric addition with superposition in the gravitational configuration space—and show that five layers of ontological structure (from the timeless Wheeler–DeWitt state to the irreversibility measure τ) resolve the "time in superposition" puzzle via the Page–Wootters mechanism. Schrödinger's cat is answered: 10^{26} internal degrees of freedom provide an entropy production $\Sigma \sim 10^{26}\sigma_1$, driving $\tau \to 1$ in femtoseconds—the cat observes itself.

INTRODUCTION

The measurement problem admits three aspects [18]: (i) the *preferred-basis* problem (why position, not momentum?), (ii) the *outcomes* problem (why one result, not many?), and (iii) the *timing* problem (when does the superposition end?). Decoherence [16, 17] addresses (i) through einselection and provides a practical answer to (iii), but standard formulations leave (ii) open—a density matrix that is diagonal "for all practical purposes" (FAPP) remains a proper-improper ambiguity.

Penrose [4, 5] confronted this problem seriously and proposed that gravity resolves it. His argument proceeds through five steps: (P1) quantum mechanics permits superpositions of geometries; (P2) two distinct mass distributions produce distinct spacetime metrics $g_{ab}^{(1)}$ and $g_{ab}^{(2)}$; (P3) therefore a spatial superposition of a massive object is simultaneously a superposition of spacetimes; (P4) such a superposition is "ill-defined" because one cannot identify points between two different manifolds, rendering the time-translation operator and hence $i\hbar\partial_t|\psi\rangle$ meaningless; (P5) therefore gravity causes an objective, non-unitary collapse at rate $\Lambda = E_G/\hbar$.

We agree with P1–P3. The problem lies in P4. Penrose's argument conflates two operations: *pointwise addition of tensor fields* on a single manifold (which requires a point identification) and *superposition in the gravitational configuration space* (which does not). Kabel *et al.* [7] made this precise: pointwise identification of manifolds is neither necessary nor meaningful for quantum gravity—"identification is pointless." Superpositions of geometries live in the Hilbert space $\mathcal{H}_{\text{geom}}$ built over Wheeler's superspace [8], where each point represents an *entire* 3-geometry. The superposition $\alpha|g^{(1)}\rangle + \beta|g^{(2)}\rangle$ is well-defined as a vector in this Hilbert space—no point identification between manifolds is needed, and $i\hbar\partial_t$ is not the relevant evolution operator (see Sec.).

This category error—from configuration-space superposition (valid) to pointwise metric superposition (invalid)—is the logical gap in the P3 → P4 step. Once removed, P5 does not follow: one needs no new collapse law.

We take a different path. In a companion paper [1], we established that the Petz recovery map [11, 12]—the unique Bayesian retrodiction functor [13]—connects quantum erasure, error correction, and thermodynamic irreversibility through the temporal asymmetry parameter $\tau = 1 - F(\rho, \mathcal{R}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho)))$. In a second companion [2], we showed that gravitational entropy production takes the form $\Sigma_{\text{grav}} = 2\ln Q = -\ln(-g_{00})$, where $Q = 1/\sqrt{-g_{00}}$ is the inverse Khronon lapse.

Here we demonstrate that collapse *is* Petz recovery failure. No new physics is needed—the existing structure of quantum information theory, when applied to many-body systems in gravitational backgrounds, produces collapse as a theorem. We establish three results:

Theorem 1 (Extensive Σ): For an N-body system, $\Sigma \geq N\sigma_{\min}$, giving $F \leq \exp(-N\sigma_{\min}/2)$ and $\tau \to 1$ for $N \gg 1$.

Theorem 2 (Σ–E_G connection): The Penrose collapse energy equals $E_G = (c^4/32\pi G)\int d^3x\ \|\nabla(\Delta\Sigma)\|^2$, linking OR to a gradient instability of the Σ field.

Theorem 3 (Information-geometric $K(Q)$): The ghost-condensation kinetic function $K(Q) = \mu^2(Q-1)^2$ is the regularized QRE of coherent-state excitations, with $\mu^2 = 2\pi$ fixed by the de Sitter temperature.

FRAMEWORK

We recall the essential ingredients from Refs. [1, 2]. The Petz recovery map for channel $\mathcal{N}$ and reference state σ is [11]

$$\mathcal{R}_{\sigma,\mathcal{N}}(\rho) = \sigma^{1/2}\,\mathcal{N}^\dagger\big(\mathcal{N}(\sigma)^{-1/2}\,\rho\,\mathcal{N}(\sigma)^{-1/2}\big)\,\sigma^{1/2}. \quad (1)$$

Parzygnat and Buscemi [13] proved that $\mathcal{R}$ is the *unique* retrodiction functor satisfying Bayesian consistency, normalization, and the correct classical limit. The temporal asymmetry parameter

$$\tau = 1 - F\big(\rho,\, \mathcal{R}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho))\big) \quad (2)$$

vanishes if and only if the channel is reversible for that state. The equivalence chain established in Ref. [1] reads

$$\tau = 0 \iff \Sigma = 0 \iff I(A;E|B) = 0 \iff \text{QMC}, \quad (3)$$

where Σ is the entropy production, $I(A;E|B)$ is the conditional mutual information of the Stinespring dilation, and QMC denotes the quantum Markov condition [14]. The universal recovery bound [15] gives

$$F \geq \exp(-\Sigma/2). \quad (4)$$

On a static gravitational background, Paper 2 [2] established

$$\Sigma_{\text{grav}} = 2\ln Q = -\ln(-g_{00}), \quad (5)$$

where $Q = 1/\sqrt{-g_{00}}$ is the inverse Khronon lapse—the local gravitational "clock-speed ratio." This identification turns the metric component g_{00} into an entropy production: stronger gravitational fields produce more Σ, making retrodiction harder.

ONTOLOGICAL STRUCTURE

The framework organizes into five levels of description, each emergent from the one above. We state them and identify which are theorems, which are standard results, and which remain interpretive.

Level 0 (Timeless).—The fundamental state $|\Psi\rangle \in \mathcal{H}_{\text{matter}} \otimes \mathcal{H}_{\text{geom}}$ satisfies the Wheeler–DeWitt constraint $\hat{H}|\Psi\rangle = 0$ [8]. No external time parameter exists. This is the ontic level; it is *not* a superposition "of different times."

Level 1 (Observer-relative).—A subsystem observer accesses

$$\rho_{\text{matter}} = \mathrm{Tr}_{\text{geom}}\big[|\Psi\rangle\langle\Psi|\big], \quad (6)$$

the reduced state obtained by tracing out gravitational degrees of freedom. This partial trace is the physical origin of $\Sigma > 0$: no subsystem can access the full correlations.

Level 2 (Branches).—Decoherence by the gravitational environment [16, 19] produces $\rho_{\text{matter}} \approx \sum_i p_i \rho_i$, an approximate mixture. Each ρ_i defines a branch. The preferred basis is selected by einselection—position, because gravity couples to mass-energy density. This is a standard result in decoherence theory.

Level 3 (Effective spacetime).—Within each branch, the Khronon field ϕ serves as a Page–Wootters clock [9]. The conditional state

$$|\psi(t)\rangle = \frac{\langle \phi = t|\Psi\rangle}{\sqrt{P(t)}} \quad (7)$$

evolves unitarily with respect to t, and the lapse $Q = 1/N_\phi$ determines $\Sigma_{\text{grav}} = 2\ln Q$ [2].

Level 4 (Irreversibility).—The temporal asymmetry parameter $\tau = 1-F$ measures how much information the subsystem observer has lost. Theorem 1 (below) shows that $\tau \to 1$ exponentially in N for macroscopic bodies.

Time is not "in superposition."—The five-level structure resolves the puzzle that Penrose raised: if geometries are superposed, "whose time do we use?" The answer: at Level 0, there is no time at all—the Wheeler–DeWitt equation is timeless. At Level 3, each decohered branch has a well-defined Khronon clock; time is perfectly sharp within a branch. The superposition exists at Level 0, but time emerges only at Level 3; these two statements are not in conflict. The Page–Wootters mechanism [9, 10] makes this precise: the "problem" of time in superposition dissolves because time is not fundamental—it is a conditional correlation.

EXTENSIVE ENTROPY PRODUCTION

The first result addresses the measurement problem directly: why do macroscopic objects always appear classical?

Theorem 1 (Extensive Σ and macroscopic classicality)**.** *Consider an N-body system $\rho_{1\ldots N}$ evolving under a channel $\mathcal{N} = \mathcal{N}_1 \otimes \mathcal{N}_2 \otimes \cdots \otimes \mathcal{N}_N$, where each $\mathcal{N}_k$ acts on subsystem k. Let $\sigma_{\min} = \min_k \Sigma_k$ denote the minimum per-subsystem entropy production. Then:*

$$\Sigma_{\text{total}} \;\geq\; N\,\sigma_{\min}. \quad (8)$$

Consequently, the recovery fidelity satisfies

$$F_{\text{total}} \;\leq\; \exp(-N\,\sigma_{\min}/2), \quad (9)$$

and $\tau \to 1$ exponentially in N.

Proof sketch.—The proof uses the data-processing inequality (DPI) for quantum relative entropy in three steps.

Step 1 (Per-subsystem bound). For each subsystem k, the coherence [22] satisfies $C(\rho_k) \geq C(\mathcal{N}_k(\rho_k))$ by DPI applied to the dephasing channel. The difference $\Sigma_k = D(\rho_k\|\sigma_k) - D(\mathcal{N}_k(\rho_k)\|\mathcal{N}_k(\sigma_k)) \geq 0$ is the per-subsystem entropy production.

Step 2 (Additivity under tensor product channels). For $\mathcal{N} = \bigotimes_k \mathcal{N}_k$ and $\sigma = \bigotimes_k \sigma_k$, the relative entropy is additive: $D(\rho_{1\cdots N}\|\sigma_{1\cdots N}) = \sum_k D(\rho_k\|\sigma_k) + I(\rho_{1\cdots N})$, where I denotes the total correlation. Since $\mathcal{N}$ is a product channel, the total correlation can only decrease: $I(\mathcal{N}(\rho)) \leq I(\rho)$. Therefore

$$\Sigma_{\text{total}} = \sum_{k=1}^{N} \Sigma_k \ \geq \ N\,\sigma_{\min}. \tag{10}$$

Step 3 (Fidelity bound). Applying (4): $F_{\text{total}} \leq \exp(-\Sigma_{\text{total}}/2) \leq \exp(-N\sigma_{\min}/2)$. Since $\sigma_{\min} > 0$ for any open subsystem, $F \to 0$ and $\tau = 1 - F \to 1$ as $N \to \infty$. □

The full proof, including the case of correlated noise, appears in the Supplemental Material [30]. The essential point survives: as long as each subsystem contributes a positive—even tiny—σ_k, the total Σ grows linearly in N, and the recovery fidelity is exponentially suppressed.

Corollary 2 (Schrödinger's cat). *A cat has $N \sim 10^{26}$ internal degrees of freedom. Each interacts with the local gravitational field through the Pikovski Hamiltonian [19] $H = H_{\text{int}}(1 + \Phi(\hat{x})/c^2)$, producing per-mode entropy $\sigma_k \sim (\Delta E_k\,\Delta\Phi\,t)^2/(\hbar c^2)^2$. Even for $\sigma_k \sim 10^{-40}$ (a single vibrational mode at room temperature over one femtosecond), the total $\Sigma \sim 10^{26} \times 10^{-40} = 10^{-14}$ already gives $F \leq e^{-10^{-14}/2} \approx 1 - 5\times 10^{-15}$ at $t \sim 1$ fs. At $t \sim 10^{-7}$ s, $\sigma_k \sim 10^{-26}$, giving $\Sigma \sim 1$ and $\tau \approx 0.4$. By $t \sim 10^{-4}$ s, $\Sigma \gg 1$ and $\tau \to 1$.*

The cat does not need an external observer. Its own 10^{26} internal degrees of freedom produce enough entropy to drive $\tau \to 1$—each atom "observes" the superposition through gravitational time dilation. The role of the "observer" is played by the extensive internal environment.

THE Σ–E_G CONNECTION

Theorem 3 (Σ–E_G connection). *Let $|\psi_1\rangle$ and $|\psi_2\rangle$ be two branches of a spatial superposition, with corresponding mass distributions $\rho_1(\mathbf{x})$ and $\rho_2(\mathbf{x})$ generating Newtonian potentials Φ_1 and Φ_2. Define the entropy production difference $\Delta\Sigma(\mathbf{x}) = \Sigma_{\text{grav}}[\Phi_1(\mathbf{x})] - \Sigma_{\text{grav}}[\Phi_2(\mathbf{x})] = -\ln(-g_{00}^{(1)}) + \ln(-g_{00}^{(2)})$. Then the Diósi–Penrose gravitational self-energy [4, 6]*

$$E_G = G\int d^3\mathbf{x}\,d^3\mathbf{x}'\ \frac{\delta\rho(\mathbf{x})\,\delta\rho(\mathbf{x}')}{|\mathbf{x}-\mathbf{x}'|} \tag{11}$$

(where $\delta\rho = \rho_1 - \rho_2$) admits the representation

$$\boxed{E_G = \frac{c^4}{32\pi G}\int d^3\mathbf{x}\ \|\nabla(\Delta\Sigma)\|^2.} \tag{12}$$

Proof.—In the weak-field limit, $g_{00} = -(1+2\Phi/c^2)$ and $\Sigma_{\text{grav}} = -\ln(-g_{00}) \approx -2\Phi/c^2$ to leading order. Hence $\Delta\Sigma \approx -2(\Phi_1 - \Phi_2)/c^2$ and

$$\nabla(\Delta\Sigma) = -\frac{2}{c^2}\nabla(\Phi_1 - \Phi_2) = -\frac{2}{c^2}\nabla\delta\Phi. \tag{13}$$

Substituting into (12):

$$\begin{aligned}\frac{c^4}{32\pi G}\int d^3x\ |\nabla(\Delta\Sigma)|^2 &= \frac{c^4}{32\pi G}\cdot\frac{4}{c^4}\int d^3x\ |\nabla\delta\Phi|^2\\ &= \frac{1}{8\pi G}\int d^3x\ |\nabla\delta\Phi|^2.\end{aligned} \tag{14}$$

By the Poisson equation $\nabla^2\delta\Phi = 4\pi G\,\delta\rho$ and integration by parts (assuming fields vanish at infinity),

$$\frac{1}{8\pi G}\int |\nabla\delta\Phi|^2\ d^3x = -\frac{1}{2}\int \delta\rho\ \delta\Phi\ d^3x = E_G. \tag{15}$$

The last equality uses $\delta\Phi(\mathbf{x}) = -G\int d^3x'\,\delta\rho(\mathbf{x}')/|\mathbf{x}-\mathbf{x}'|$, giving the standard Penrose expression (11). □

Equation (12) reinterprets E_G as a gradient energy of the Σ field. Collapse occurs when the entropy-production landscape has sharp gradients—when different branches of the superposition "see" different amounts of irreversibility.

Equivalence principle and E_G.—If $\Delta\Sigma$ = const everywhere (uniform redshift difference), then $\nabla(\Delta\Sigma) = 0$ and $E_G = 0$. A uniform gravitational field cannot cause collapse—consistent with the equivalence principle. Collapse requires *tidal* (non-uniform) gravitational effects, which are precisely the effects that $\nabla(\Delta\Sigma) \neq 0$ detects.

Collapse timescale.—Penrose's collapse time $t_{\text{DP}} = \hbar/E_G$ becomes

$$t_{\text{DP}} = \frac{32\pi G\hbar}{c^4\int d^3x\ \|\nabla(\Delta\Sigma)\|^2}, \tag{16}$$

which is long when Σ is smooth (microscopic superpositions) and short when Σ has large gradients (macroscopic superpositions of distinct mass configurations).

INFORMATION-GEOMETRIC ORIGIN OF $K(Q)$

The third result connects the Σ framework to the kinetic structure that governs ghost condensation [21] and appears in the weak-field dark matter phenomenology of Paper 3 [3].

Theorem 4 (Coherent-state QRE and ghost condensation). *Let $|\delta\rangle$ be a coherent-state excitation of the*

Khronon field with $Q = 1 + \delta$, and let $|0\rangle$ denote the ground state ($Q = 1$, Minkowski). The regularized quantum relative entropy

$$D_{\mathrm{reg}}(|\delta\rangle \| |0\rangle) = \lim_{\epsilon \to 0} \left[D(\rho_\delta \| \rho_0) - D_{\mathrm{div}}(\epsilon) \right] \tag{17}$$

(where D_{div} removes the UV-divergent vacuum contribution) satisfies

$$\boxed{D_{\mathrm{reg}}(|\delta\rangle \| |0\rangle) = (Q-1)^2.} \tag{18}$$

The kinetic function of the Khronon field is therefore

$$K(Q) = \mu^2 \, D_{\mathrm{reg}}(|\delta\rangle \| |0\rangle) = \mu^2 (Q-1)^2, \tag{19}$$

where $\mu^2 = T_{\mathrm{dS}}/(2\hbar)$ is fixed by the de Sitter temperature $T_{\mathrm{dS}} = H_0 \hbar/(2\pi k_{\mathrm{B}} c)$, giving $\mu = H_0/c$ in natural units [2, 3].

Proof sketch.—For coherent states $|\alpha\rangle$ and $|\beta\rangle$, the overlap is $\langle\beta|\alpha\rangle = \exp(-|\alpha - \beta|^2/2 + i\,\mathrm{Im}(\beta^*\alpha))$. The quantum fidelity is $F(|\alpha\rangle, |\beta\rangle) = \exp(-|\alpha-\beta|^2/2)$, giving $-\ln F^2 = |\alpha - \beta|^2$.

Identifying $\alpha = \delta$ (the Khronon excitation) and $\beta = 0$ (vacuum), we obtain $-\ln F^2 = |\delta|^2 = (Q-1)^2$. Since the Pinsker bound saturates for coherent states—they are Gaussian, and all Rényi divergences coincide up to normalizations for Gaussian states—the regularized QRE equals $(Q-1)^2$.

The quadratic form is *unique* among Gaussian states: the coherent-state QRE is always $|\alpha - \beta|^2$ regardless of the number of modes. Non-Gaussianity (UV effects) produces higher-order corrections $\sim (Q-1)^4$, corresponding to the DBI modification of $K(Q)$. The full proof appears in [30]. □

Ghost condensation from information geometry.—At the ghost-condensation point $Q = Q_0 \approx 1$, we have $K'(Q_0) = 0$ and $K''(Q_0) = 2\mu^2 > 0$. This is the condition identified in Ref. [3] (Theorem 1 therein) as the origin of $c_s^2 = 0$ for the dark-matter-like perturbations. Theorem 4 provides its information-geometric origin: ghost condensation occurs at the *minimum distinguishability* between the Khronon configuration and vacuum. The field "condenses" at the state closest to vacuum in the relative-entropy sense.

Uniqueness.—The quadratic form $K(Q) = \mu^2(Q-1)^2$ is the unique kinetic function that: (i) has a ghost-condensation point at $Q_0 = 1$ ($K'(1) = 0$); (ii) arises from a Gaussian (coherent-state) QRE; (iii) recovers the correct de Sitter thermodynamics when $\mu = H_0/c$. Non-Gaussian corrections are suppressed at low energies by the central limit theorem applied to the many IR modes of the Khronon field.

DISCUSSION

Precise divergence from Penrose.—We agree with Penrose on P1–P3: quantum mechanics permits geometry superpositions, distinct masses produce distinct metrics, and a massive superposition is simultaneously a superposition of spacetimes. We disagree on P4: the "ill-definedness" argument commits a category error, confusing pointwise metric addition (which requires identifying points between manifolds) with superposition in the gravitational configuration space (which does not). Kabel *et al.* [7] demonstrated that diffeomorphism-invariant quantum gravity renders point identification meaningless—the Hilbert space is built over equivalence classes of 3-geometries, not over labeled point sets. Consequently P5 does not follow: we need no new dynamical collapse law. Our framework reproduces the *phenomenology* of Penrose's E_G (Theorem 3) as a derived quantity—the gradient energy of the Σ field—while remaining within standard quantum mechanics.

Epistemological status.—Theorems 1 and 3 are mathematical results within standard quantum mechanics plus Newtonian gravity. The five-level ontological structure (Sec.) is an interpretive framework—it organizes known physics but does not itself constitute a new prediction. We mark this distinction clearly: Theorems 1–3 are provable; the Page–Wootters clock interpretation of the Khronon is a *proposal*; and the complete derivation of Level 0 from a UV-complete quantum gravity remains open. We do not resolve whether other branches "cease to exist." What Theorem 1 establishes is that no physical protocol can recover a macroscopic superposition once $\Sigma \gg 1$.

Experimental distinction from Penrose OR.—The two frameworks make a sharp, testable prediction about *recoherence*. Penrose OR predicts that collapse is a genuine, irreversible, non-unitary process: once a superposition decays via $\Lambda = E_G/\hbar$, it cannot be restored even in principle. Our framework predicts that collapse is *operationally* irreversible (exponentially hard to undo) but *in-principle* reversible: if all N subsystems and their gravitational environment could be coherently controlled, the Petz map would succeed and the superposition would recohere.

The key test is therefore a *recoherence experiment*: prepare a mesoscopic superposition, let it decohere ($\tau \to 1$), then attempt to reverse the decoherence channel. Penrose OR predicts zero recoherence probability. We predict $F_{\mathrm{recoh}} = \exp(-\Sigma/2)$—exponentially small but nonzero. Current experiments cannot reach the required control, but the logical structure differs.

Existing data already constrain the Diósi–Penrose (DP) model. Underground experiments at Gran Sasso [26] excluded the parameter-free version of the DP collapse model via absence of anomalous radiation from spontaneous collapse events. The 2026 matter-wave interference experiment by Pedalino *et al.* [27] demonstrated coherence for molecules at 170 kDa, exceeding the mass range where the simplest DP predictions would require visible decoherence. Both results are consistent with our framework, which predicts no anomalous decoherence beyond that from the standard Σ mechanism.

Schrödinger's cat without an observer.—The cat's 10^{26} atoms interact with the local gravitational field through position-dependent time dilation. Each atom contributes $\sigma_k > 0$ to the total entropy production. By Theorem 1, $\Sigma \geq 10^{26}\sigma_{\min}$, and no recovery is possible. The cat does not need a physicist to open the box. It classicalizes through its own internal complexity—the atoms are the observer.

Connection to the Weyl curvature hypothesis.—Penrose's Weyl curvature hypothesis [23] states that $C_{abcd} = 0$ at the Big Bang. In the Σ language: $C_{abcd} = 0$ corresponds to $\Sigma_{\rm grav} = 0$ (conformally flat spacetime), which by the equivalence chain (3) gives $\tau = 0$. The initial singularity has zero temporal asymmetry—no arrow of time, no collapse, full quantum coherence. The growth of Σ (via gravitational clumping and structure formation) produces the cosmological arrow of time. This matches Penrose's picture but arises here from information theory rather than an independent postulate.

Experimental predictions.—Pikovski-type experiments [19] that place massive objects in superposition probe $\Sigma_{\rm grav} \sim N_{\rm int}$. The Bose–Marletto–Vedral (BMV) experiment [24] targets $\tau \sim 0.004$. Our framework predicts a sharp transition: $\tau \ll 1$ for current molecule interferometry ($N \sim 10^3$ atoms [25]), but $\tau \to 1$ for proposed micro-mechanical oscillators ($N \sim 10^{14}$). Any deviation from the baseline $\tau(t) = 1 - \exp(-\Sigma(t)/2)$ signals physics beyond standard quantum mechanics.

Open: Born rule.—The Born rule is assumed throughout. Parzygnat and Buscemi's retrodiction framework [13] suggests that the Born rule may be derivable from the Bayesian structure of the Petz map. If so, the entire measurement problem—preferred basis, timing, and probabilities—would reduce to properties of Petz recovery. Oldofredi and Calosi [28] showed that relational quantum mechanics faces tension with the PBR theorem unless supplemented by an information-loss mechanism—Petz recovery failure may supply exactly this. We leave this as an open direction.

Open: full Level 0 derivation.—The five-level ontology (Sec.) assumes the Wheeler–DeWitt constraint at Level 0. Deriving this from a UV-complete theory of quantum gravity—and proving that the Page–Wootters conditional state reproduces the Khronon structure at Level 3—remains an open problem. We regard the present framework as a consistency argument: known structures, assembled correctly, produce collapse without new postulates.

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[30] See Supplemental Material for full proofs of Theorems 1–4, the correlated-noise generalization, the coherent-state QRE calculation, and numerical estimates for the Schrödinger's cat scenario.